

8/8/26

Activity CO-3

Hough Transform

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1. given points $P_1 = (5, 4)$
 $P_2 = (7, 6)$

$$C_1 = (2, 4)$$

$$C_2 = (4, 3)$$

First C_1 and P_1

$$\begin{array}{cc} x_1 & y_1 \\ C_1 & = (2, 4) \end{array}$$

$$P_1 = (5, 4)$$

$$\begin{array}{cc} x_2 & y_2 \end{array}$$

$$d = \sqrt{(5-2)^2 + (4-4)^2}$$

$$= \sqrt{3^2 + 0^2} = \sqrt{9} = 3 \quad \boxed{d=3}$$

 C_1 and P_2

$$C_1 = (2, 4)$$

$$P_2 = (7, 6)$$

$$d = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{5^2 + 2^2} = \sqrt{25+4} = \sqrt{29}$$

$$\boxed{d = \sqrt{29}}$$

 C_2 and P_1

$$(4, 3)$$

$$(5, 4)$$

$$d = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \boxed{d = \sqrt{2}}$$

 C_2 and P_2

$$d = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{3^2 + 3^2} = \sqrt{9+9} = \sqrt{18}$$

$$\boxed{d = \sqrt{18}}$$

∴ Therefore No centre gives the same radius

2. given $P(8,6)$ x_2, y_2
 $C(4,3)$ x_1, y_1

$$\text{Radius} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{(4)^2 + (3)^2}$$

$$= \sqrt{16+9}$$

$$= \sqrt{25} = 5$$

$$\boxed{r=5}$$

3. given $C=(5,4)$ x_1, y_1
 $P=(8,8)$ x_2, y_2

$$\text{Radius} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{(3)^2 + (4)^2} = \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5$$

$$\boxed{r=5}$$